```
/* Name-Antara Mangle
  /PRN-25070521199/
  SECTION-D-I
  Value Added Problem-2 */
#include <stdio.h>
#include <string.h>

#define MAX 1000000

char stack[MAX];
int top = -1;

// Push function
void push(char c) {
   stack[++top] = c;
}

// Pop function
char pop() {
   if (top == -1)
      return '\0';
   return stack[top--];
}

// Check if matching pair
int isMatchingPair(char open, char close) {
   if (open == '(' && close == ')') return 1;
   if (open == '{' && close == '}') return 1;
   if (open == '[' && close == ']') return 1;
```

```
    return 0;
}

int isBalanced(char *s) {
    for (int i = 0; s[i] != '\0'; i++) {

        // If opening bracket, push
        if (s[i] == '(' || s[i] == '{' || s[i] == '[') {
            push(s[i]);
        }
        // If closing bracket
        else if (s[i] == ')' || s[i] == '}' || s[i] == ']') {

            if (top == -1)  // No matching opening
                return 0;

            char topElement = pop();

            if (!isMatchingPair(topElement, s[i]))
                return 0;
        }
    }

    // If stack is empty → balanced
    return (top == -1);
}

int main() {
    printf("Name :Antara Mangle\n");
```

```
    printf("PRN :25070521199\n");
    printf("Section :D-1\n");
    printf("Value Added Problem : 2\n");
    printf("--------------------------\n");
    char s[MAX];

    printf("Enter expression: ");
    scanf("%s", s);

    if (isBalanced(s))
        printf("true\n");
    else
        printf("false\n");

    return 0;
}
```

main.c

```
/* Name-Antara Mangle
   /PRN-25070521199/
   SECTION-D-I
   Value Added Problem-2 */
#include <stdio.h>
#include <string.h>

#define MAX 1000000

char stack[MAX];
int top = -1;
```

Output

```
Name :Antara Mangle
PRN :25070521199
Section :D-1
Value Added Problem : 2
--------------------------
Enter expression: {([])}
true


=== Code Execution Successful ===
```